Sai Jagarlamudi

ai25mtech14002@iith.ac.in | jagarlamudisai0602@gmail.com | +91 8639070118

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.32
Bhashyam Junior College	Class XII (Board of Intermediate Education, AP)	2020	95.20%
Bhashyam High School	Class X (Board of Secondary Education, AP)	2018	10.00%

PROJECTS

RadXAI: Mitigating Hallucination in Radiology VLMs via RAG | PyTorch, LLaVA-1.5-7B, LoRA, FAISS

- Built a visual retrieval-augmented generation pipeline to mitigate hallucinations in medical vision-language models,fine-tuning LLaVA-1.5-7B with LoRA on CheXpert Plus chest X-ray reports.
- Designed test vqa, a hallucination benchmark with automatic clinical entity extraction and sibling-safe negative sampling, evaluated across CheXpert, MIMIC-CXR, and IU X-Ray (>17K questions).
- Improved finding-level precision from 0.55 to 0.84 and reduced hallucination rate from 45.1% to 15.6% by grounding generation in CheXzero/FAISS-retrieved reference cases instead of the query image alone.

dualmemory | *Python, SQLite, FAISS, PyPI*

- Built and published an open-source, pip-installable semantic +episodic memory library for LLM agents.
- Designed a dual-store architecture pairing timestamped episodic events with durable semantic facts, where every memory is traceable to its source and outdated facts are superseded instead of overwritten.
- Added pluggable embedding, LLM, and storage backends for a model-agnostic core, plus a consolidation step that turns recurring events into lasting facts.
- Benchmarking on LoCoMo and LongMemEval against no-memory / episodic / semantic / hybrid baselines; core build in progress.

ManimForge:Agentic Prompt-to-Video Generation | Python, Gemini API, Manim CE, Docker

- Built an end-to-end agentic pipeline converting natural language prompts into rendered educational animations, using a constrained system prompt to generate executable Manim Community Edition code via the Gemini API.
- Implemented static validation (AST parsing) before execution, sandboxed subprocess-based rendering with timeout controls, and an error-classification and self-correction retry loop to recover from generation failures.
- Containerized the rendering environment with Docker to isolate LLM-generated code execution, addressing arbitrary code execution risk inherent to running untrusted generated scripts.

Sorting Algorithm Visualizer | HTML, CSS, JavaScript

- An interactive web application implementing **5** sorting algorithms**.**Quick, Merge, Bubble, Selection, and Insertion Sort with step-by-step visualizations.
- Implemented asynchronous sorting animations using JavaScript Promises, async/await, setTimeout, and the Web Animations API to coordinate comparisons and element swaps.
- Developed real-time comparison and swap tracking, configurable animation speed, custom array input, and randomized array generation for interactive experimentation.

PUBLICATIONS

RadXAI: Probing and Mitigating Hallucination in Radiology VLMs with Visual Retrieval Augmented Generation and Policy Optimization

- **Accepted as a Poster** at the **WiML Workshop @ NeurIPS 2026**
- Also submitted to the **Forty-First AAAI Conference on Artificial Intelligence (AAAI-27)** -- *Under Review*

SKILLS

Languages: Python, C, C++, JavaScript

ML/DL Frameworks: PyTorch, Hugging Face Transformers, Unsloth, PEFT

ML/DL Concepts: Retrieval-Augmented Generation (RAG), Vector Search, Model Fine-tuning, Semantic/Episodic Memory

Frameworks & Libraries: NumPy, Pandas, Matplotlib, FAISS, Pydantic

Tools & Platforms: Git, GitHub, Linux, Docker, Jupyter Notebook, Google Colab, Postman

APIs: Gemini API, OpenAI API, REST APIs

RELEVANT COURSES

Advanced Mathematics: Probability, Convex Optimization, Matrix Theory.

Core: Fundamentals of Machine Learning, Deep Learning.

POSITION OF RESPONSIBILITY

Teaching Assistant, Multi-Agent Systems, IITH

Assisted 100+ students by clarifying course concepts and evaluating assignments.